Row and Column Vectors

Multiplication right to left

$\mathbb{R}^n = \left\{ \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}, a_i \in \mathbb{R} \right\}$ $\quad \mathbb{R}^{m\times n} = \left\{ A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ a_{m1} & \cdots & \cdots & a_{mn} \end{bmatrix} ; a_{ij} \in \mathbb{R} \right\}$

(inputs) (outputs)

$\mathbb{R}^n \cong \mathbb{R}^{n\times 1}$ column vector is 1 column by n rows (vertical)

$(\mathbb{R}^n)' := \mathbb{R}^{1\times n}$ row vector 1 row by n cols (vertical)

$\left\{ [b_1 \cdots b_n] ; b_i \in \mathbb{R} \right\}$

Matrix Multiplication:

$\mathbb{R}^{m\times p} \times \mathbb{R}^{p\times n} \longrightarrow \mathbb{R}^{m\times n}$ $\qquad (AB)_{ij} = \sum_{k=1}^{p} A_{ik} B_{kj}$ for $\begin{matrix} 1\le i\le m \\ 1\le j\le n \end{matrix}$

$A \cdot B = C$

$(m\times p)\ (p\times n) \to (m\times n)$

↳ Associative $ABC = (AB)C = A(BC)$

↳ Not usually Commutative $AB \neq BA$ often

↳ Distributes across addition $A(B+C) = AB + AC$

$(A+B)C = AC + BC$

↳ Commutes with scale multiplication $A(cB) = c(AB) = (cA)B$

Transpose $\mathbb{R}^{m\times n} \to \mathbb{R}^{n\times m}$

$A \mapsto A'$ where $(A')_{ij} = (A)_{ji}$

Flip rows and columns

$(AB)' = B' \cdot A'$

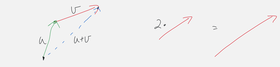
$(m\times n)' = n\times m$

A vector space V has scalar multiplication and additive group $(V, +, 0)$ satisfying $a+b = b+a$

$c\in\mathbb{R}$ { Stretch by $|c|$ / Flip if $c<0$ / Zero if $c=0$ }

Elements of vector spaces = "vectors" = things we can add, subtract, and scale.

$ab \overset{?}{=} ba$



In general a vector space has no notions of "distance, angle, length, size"

Eg. $\mathbb{R}[x] = \{ g(x) = a_k x^k + \cdots + a_1 x + a_0 ; \ k\in\mathbb{N}, a_i \in \mathbb{R} \}$ (P)

↳ "What's the length of $x^2 - 5$?"

↳ "What's the angle between $5x^7 - 4$ and $2x+1$?"

A vector space that has distances and angles is an inner product space.

For $\mathbb{R}^n$, we have dot product

$\vec{u}\cdot\vec{v} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \sum_{i=1}^{n} u_i v_i = u_1 v_1 + \cdots + u_n v_n$

Geometry: Angle of $\vec{u}$ and $\vec{v}$ $= \arccos\left(\frac{\vec{u}\cdot\vec{v}}{\text{mag mag}} \right) = \arccos(\hat{u}\cdot\hat{v})$

Unit vector of $\vec{u}$ is $\hat{u} = \frac{1}{\|\vec{u}\|}\vec{u}$ (direction) (size)

Length or Norm of $\vec{u}$ is $\|\vec{u}\| = \sqrt{\vec{u}\cdot\vec{u}}$ — derived directly from the Pythagorean theorem

The **dot product** of two vectors $\vec{a}, \vec{b} \in \mathbb{R}^n$ is defined as:

$\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \cos\theta$

Any vector is defined uniquely by its direction and norm

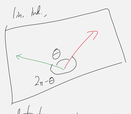
$\vec{u} = L \cdot \hat{u}$ $\quad L\in\mathbb{R}, L\ge 0, \ L = \|\vec{u}\|$

$\hat{u}\in\mathbb{R}^n, \ \|\hat{u}\| = 1$

(if $L=0$, $\hat{u}$ can be anything, but if $L\neq 0$, $\vec{u}$ is defined uniquely by L and $\hat{u}$)

$\vec{u} = L\hat{u}$, $\vec{v} = M\cdot\hat{v}$ $\quad \cos\theta = (\hat{u}\cdot\hat{v}) = \begin{cases} >0 & (\text{Acute}) \ 0\le\theta<\frac{\pi}{2} \\ =0 & (\text{Right}) \ \theta = \frac{\pi}{2} \\ <0 & (\text{Obtuse}) \ \frac{\pi}{2}<\theta\le\pi \end{cases}$

If u, v are in $\mathbb{R}^3$,



Dot Products detect angle with sign

Dot Products Compute matrix multiplication

(Matrix Multiplication, Linear Algebra) $\leftrightarrow$ (Dot Products, Geometry)

$(\mathbb{R}^{1\times 1} \cong \mathbb{R})$

$\vec{u}\cdot\vec{v} = \sum_{i=1}^{n} u_i v_i = \begin{bmatrix} u_1 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = u'v \quad \hookrightarrow \ u'v = v'u$

$(AB)_{ij} = \sum_{k=1}^{p} A_{ik} B_{kj} = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1p} \end{bmatrix} \begin{bmatrix} B_{1j} \\ B_{2j} \\ \vdots \\ B_{pp} \end{bmatrix}$ (j^{th} column)

(i^{th} Row) $\begin{bmatrix} A_{11} & \cdots & A_{1p} \\ A_{21} & \cdots & A_{2p} \\ A_{i1} & A_{i2} & \cdots & A_{ip} \\ A_{m1} & \cdots & A_{mp} \end{bmatrix}$ $\quad \begin{bmatrix} B_{11} & B_{12} & \cdots & B_{1n} \\ B_{21} & B_{22} & & B_{2n} \\ \vdots & & & \vdots \\ B_{p1} & B_{p2} & \cdots & B_{pn} \end{bmatrix}$

$(AB)_{ij} = A_{i\text{th row}} \cdot B_{j\text{th column}}$

$AB = \begin{bmatrix} - & R_1 & - \\ & \vdots & \\ - & R_m & - \end{bmatrix} \begin{bmatrix} | & & | \\ C_1 & \cdots & C_n \\ | & & | \end{bmatrix} = \begin{bmatrix} R_1\cdot C_1 & R_1\cdot C_2 & \cdots & R_1\cdot C_n \\ R_2\cdot C_1 & R_2\cdot C_2 & & \vdots \\ \vdots & & \ddots & \\ R_m\cdot C_1 & \cdots & & R_m\cdot C_n \end{bmatrix}$

A B AB

So, each entry of AB is a dot product

↕

The sign of each entry is telling us whether that Row and that Column (two arrows in $\mathbb{R}^n$) are acute or obtuse (roughly same direction) (roughly opposite direction)

$x\in\mathbb{R}^n$, $A\in\mathbb{R}^{m\times n}$

$A\vec{x} \in \mathbb{R}^m$ $\quad A\vec{x} = x_1 A_1 + x_2 A_2 + \cdots + x_n A_n$ Linear Combination of columns of A (weights that components of $A\vec{x}$ are)

$\begin{bmatrix} | & | & & | \\ A_1 & A_2 & \cdots & A_n \\ | & | & & | \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

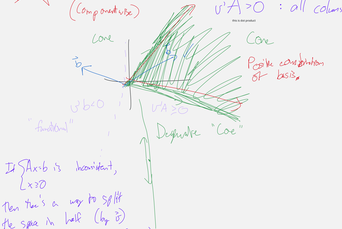
$\begin{bmatrix} x_1 a_{11} \\ x_1 a_{21} \\ \vdots \\ x_1 a_{m1} \end{bmatrix} + \begin{bmatrix} x_2 a_{12} \\ x_2 a_{22} \\ \vdots \\ x_2 a_{m2} \end{bmatrix} + \cdots + \begin{bmatrix} x_n a_{1n} \\ x_n a_{2n} \\ \vdots \\ x_n a_{mn} \end{bmatrix} = \begin{bmatrix} x_1 a_{11} + x_2 a_{12} + \cdots + x_n a_{1n} \\ \vdots \\ x_1 a_{m1} + \cdots + x_n a_{mn} \end{bmatrix} = \begin{bmatrix} R_1 \cdot x \\ \vdots \\ R_m \cdot x \end{bmatrix}$

$A\vec{x} = \begin{bmatrix} - & R_1 & - \\ & \vdots & \\ - & R_m & - \end{bmatrix} \begin{bmatrix} | \\ \vec{x} \\ | \end{bmatrix} = \begin{bmatrix} R_1 \cdot \vec{x} \\ R_2 \cdot \vec{x} \\ \vdots \\ R_m \cdot \vec{x} \end{bmatrix}$

$A\vec{x} \ge 0 \iff \vec{x}$ is "same direction" as all Rows of A:

(Componentwise) $\quad v'A \ge 0$: all columns of A are "same $\vec{x}$, $\vec{v}$" as $\vec{v}$

this is dot product



If $\exists Ax = b$ is inconsistent, $\begin{cases} x\ge 0 \end{cases}$

Then there's a way to split the space in half (by $\vec{v}$) we "$A_i \cdot \vec{v}$" ≥ 0 $(v'A)$

"$\vec{b}\cdot\vec{v}$" < 0 $(v'b)$

All the A's go one way, but $\vec{b}$ goes opposite.